

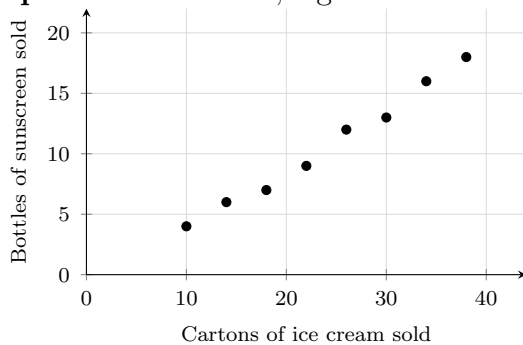
Scatter Plots and Bivariate Association — Does It Show Cause, or Only a Pattern?

Two quantities are **associated** when knowing one tells you something about the other. That is what a scatter plot can show. It cannot, by itself, show that one quantity **causes** the other.

- A **lurking variable** is a third quantity that moves both.
- **Reverse cause** means the arrow may point the other way.
- Only a study that **assigns** the treatment at random can support a cause claim.

Use the three displays below. Every problem says which one it needs.

Graph A — one town, eight summer weeks



Graph C — six students, one school night

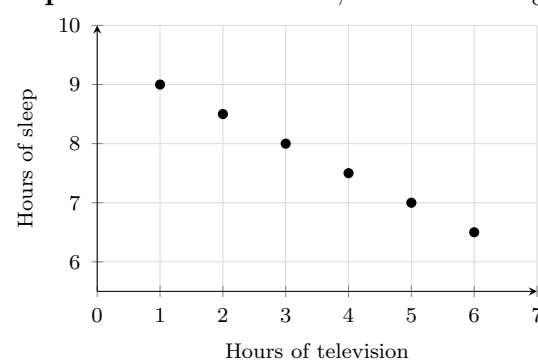


Table B — test scores of ten students who chose for themselves whether to use a tutoring app

Used the app	88	79	95	84	74
Did not use it	72	66	81	70	76

1. Graph A. Read one point off the plot: in the week when 26 cartons of ice cream were sold, how many bottles of sunscreen were sold?

Answer: _____

2. Graph A. Find and fix. Dana is checking how much the sunscreen sales climb across the summer. She compares the week with 10 cartons to the week with 38 cartons and writes: “Sunscreen went up by 28.” Dana read the wrong axis. How many more bottles of sunscreen were sold in the 38-carton week than in the 10-carton week?

Answer: _____

3. Graph A. Dana now writes: “Graph A proves that selling ice cream causes people to buy sunscreen.” Name one lurking variable that would push both sales up together, and write one sentence explaining why an association on a scatter plot cannot by itself show cause.

4. Table B. Find and fix. Ravi wants the mean score of the students who used the app. He typed every score in the table into his calculator and reported a mean of 78.5. Ravi averaged the wrong set of students. Find the correct mean score for the group that used the app.

Answer: _____

5. Table B. Now find the mean score of the group that did not use the app.

Answer: _____

6. Table B. The app’s advertisement says: “Our app raises test scores by eleven points.” Using the two means you just found, explain why this study cannot support that claim — name one way the two groups could differ apart from the app — and describe one change to how the study was run that would let it support a cause claim.

7. Graph C. Find and fix. Sam says, “Watching television makes you lose sleep,” and reports the size of the effect as 5 by subtracting the television hours. Sam used the wrong axis again. Going from the student who watched 1 hour to the student who watched 6 hours, what is the change in hours of sleep? Write it as a signed number.

Answer: _____

8. Challenge — Graph C. Rewrite Sam's headline so that it claims only what Graph C actually shows. Then describe the experiment a researcher would have to run to test whether television time really causes sleep loss — say who is assigned what, and how the assignment is made.